

Mamdouh Attia

Mobile Software Engineer [Flutter]

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EDUCATION

Cairo University - Faculty of Engineering

Cairo, Egypt

B.E. – Computer Engineering Department

2019 - 2024

- Grade : Very Good with Honors.
- Graduation Project's Grade : Excellent.

EXPERIENCE

Programming Fundamentals Instructor (Python)

Jun. 2025

Maspero Academy - Radio and Television Institute

Cairo, Egypt — Hybrid

- Delivered +20 hr Python programming basics course to CS students, creating engaging coding exercises.

Machine Learning Engineer Instructor

Aug. 2024 – Oct. 2024

DEPI Initiative, Ministry of Communications and Information Technology

New Damietta, Egypt - Hybrid

- Delivered +100 hr hands-on machine learning training sessions for +15 aspiring engineers.
- Designed & evaluated +17 Tasks & final project to cover essential maths, coding, ML concepts and tools.

Flutter Instructor

Aug. 2023 – Feb. 2024

Primalab

New Damietta, Egypt

- Taught Dart programming & Flutter essentials to +10 students, including layout, widgets & navigation.
- Guided students through +28 hands-on sessions, +10 tasks and project-based learning.

Flutter Developer Intern

Jun. 2023 – Sep. 2023

Orange Digital Center Software House | [Certificate Link](#)

Remote

- Worked with agile teams on 2 mobile app projects & 1 hackathon fintech app, meeting tight sprint deadlines
- Translated Figma designs into Flutter UI & used MVVM architecture and Bloc Cubit

FLUTTER PROJECTS

EcoSort | Flutter, Freelance, Firebase, Blync API, QR, Push Notifications

Apr. 2024 – May. 2024

- Enhanced an existing client app by adding Firebase authentication and push notifications.
- Integrated Blync API to enable real-time connection with 3 hardware sensors in recycle bins.
- Implemented QR code scanning and a point reward system for recycling activities.

Check-It | Flutter, Freelance, Barcode, Provider, Forms | [Portfolio](#)

Feb. 2024 – Jun. 2024

- Developed a mobile solution to verify product prices and combat price manipulation in retail.
- Implemented barcode scanning, real-time price comparison, and e-commerce style product database.
- Enabled users to submit complaints and track their status.

VolunTeen | Flutter, Freelance, Provider, Firebase, GPT-3 | [Portfolio](#) | [Demo Link](#)

Sep. 2023 – Jan. 2024

- Solo project for a real client: built a volunteering app targeting teenagers, featuring responsive 30–40 screens.
- Integrated AI chatbot (GPT-3), in-app video player, and third-party services.
- Implemented MVC architecture, Firebase (storage), shared preferences, and WhatsApp/URL launcher.

HyperMate | Flutter, Freelance, Wi-Fi service, ESP8266, GPT-3 | [Portfolio](#) | [Demo Link](#)

Nov. 2023 – Dec. 2023

- Solo client project: developed a stress monitoring app connecting to ESP8266 microcontroller via Wi-Fi.
- Displayed real-time GSR sensor readings and provided personalized exercise and nutrition suggestions.
- Integrated a smart chatbot for user support and recommendations, and handled the arduino firmware code.

TabCash | Flutter, ODC's Hackathon, Agile, MVC | [Repo link](#) | [Presentation link](#)

Apr. 2023 – May. 2023

- Developed a fintech app in a 2-week hackathon enabling secure and easy transactions.
- Built UI for home, wallet, statistics, and profile screens, and presented to Orange Digital Centre's board.

Reddit Clone | Flutter, MVC-S, Agile, Provider, REST APIs | [Repo Link](#) | [Demo Link](#)

Aug. 2022 – Dec. 2022

- Developed a Reddit mobile app clone as part of a 6-member team in a software house environment.
- Implemented UI components including drawers, settings, and community creation screens.
- Contributed to unit testing, mock services, documentation, and backend integration.
- Followed Agile methodology with dedicated roles for frontend, backend, DevOps, and QA.

GRADUATION PROJECT

- Predicto** | *A+ | Python, OpenCV, Pytorch, Deep Learning, Desktop, Research* | [Demo video](#) Aug. 2023 – July 2024
- Developed an open-source Python library named 'vPredicto', specialized in VFP (Video Frames Prediction).
 - Integrated the latest state-of-the-art Deep Learning models in this field.

OTHER PROJECTS

- Web-page Phishing Detection** | *Python, Machine Learning, scikit-learn* | [Repo Link](#) Apr. 2024 – May. 2024
- Addressed the phishing detection problem by training & evaluating 12 ML models with hyperparameter tuning.
 - Applied advanced evaluation metrics and ML concepts to address real-world cybersecurity threats.
- Gaming Engine + Drive Me Home** | *C++, OpenGL, OOP* | [Repo Link](#) | [Demo Link](#) Mar. 2023 – May. 2023
- Co-developed a custom game engine & a game using C++ & OpenGL, implementing shaders, textures & lighting.
 - Engine: Designed shader, texture, sampler, and material classes.
 - Game: Built collision system, post-processing shaders, menus, level-up/game-over states & multiple game objects.
- RSA E2E Encrypted Chat** | *Python, Security, Socket Programming* | [Repo Link](#) | [Explanation video](#) Apr. 2023
- Created a secure Python chat service with end-to-end RSA encryption using sockets.
- MIPS Processor** | *Verilog, Computer Architecture, FPGA Simulation, IC Design* | [Repo Link](#) Dec. 2022
- Co-designed a 5-stage pipelined MIPS processor in Verilog with hazard detection, forwarding & branch control.
 - Role: Decode stage & system integration, ensuring correct instruction flow & modular pipeline operation.
- Search Engine** | *Java, OOP, Multithreading, Web Development* | [Repo Link](#) Apr. 2022
- Co-developed a modular Java-based search engine with a web interface.
 - Implemented the web crawler module in Java, handling multithreaded crawling & URL normalization.
- OS Scheduler** | *C, Operating Systems* | [Repo Link](#) Apr. 2022
- Co-developed an OS-like scheduler implementing process scheduling algorithms in C on Linux.
- Covid Detector** | *Python, Deep Learning, TensorFlow, Keras* | [Repo Link](#) Sep. 2021
- Built a deep learning model to detect Covid-19 from X-ray images using VGG16, VGG19 & AlexNet architectures.
 - Applied advanced DL concepts and evaluated model performance on real datasets.
- Rovers Mars-station** | *C++, Data Structures & Algorithms, OOP, Event driven* | [Repo Link](#) Apr. 2021
- Co-Developed a C++ simulation system modeling Mars exploration missions.
 - Implemented custom data structures (queues, priority queues) and event-driven design.
 - Implemented modular OOP architecture for mission scheduling, rover management, and performance reporting.
- Logic Simulator** | *C++, OOP, CMU Graphics, Design Patterns, Simulation, Desktop* | [Repo Link](#) Jan. 2021
- Co-developed a C++ desktop app to design and simulate digital logic circuits using OOP & design patterns.
 - Implemented an interactive GUI with real-time simulation and drag-and-drop gate control.

SKILLS

- Programming:** C/C++ | Dart | Python | Java
- Mobile:** Flutter | Provider | Bloc/Cubit | MVC | MVVM | Clean Architecture | Hardware Integration | Streams | Shared Prefs | Firebase | Supabase | API Integration | Unit Testing | Documentation
- Software Eng:** SDLC | SOLID | OOP | Data Structures | Algorithms | Design Patterns | Clean Code | Linux | Git/GitHub (Actions / Projects) | Agile / SCRUM
- AI:** ML/DL | Computer Vision | NLP | Keras | TensorFlow | PyTorch | Azure AI

CERTIFICATES

- Career Essentials in GitHub Professional Certificate** | [Certificate Link](#) 2026
- Coursera's ML Specialization** | [Certificate Link](#) 2024
- Huawei HCIA-AI** | [Certificate](#) 2021

VOLUNTEERING

Instructor to the Staff of My City's Orphanage

Damietta, Egypt

- Microsoft Word Fundamentals Instructor Aug. 2022 – Sep. 2022
- Windows Fundamentals Instructor Aug. 2021 – Sep. 2021